ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

ALLIANCE SCHOOL OF ADVANCE COMPUTING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR 2026 -2027

Course of Independent Study (CIS) Diary

CIS Course Code	:	7CS 9281
CIS Title	:	Automated Student Performance Prediction
SUPERVISOR NAME	:	Prof. Jessy Janet Kumari
STUDENT SCHOOL	:	Alliance School of Advance Computing
NAME OF THE STUDENT (Register No)	:	1. Chalini M (2023BCSE07AED440)
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Abstract of the Project

The Automated Student Performance Prediction project uses Machine Learning and AutoML techniques to identify students who may be at academic risk. Using the UCI Student Performance dataset, the system analyzes student attributes such as study time, absences, family background, and past academic behavior. It automatically compares multiple classification algorithms, tunes their parameters, and selects a suitable model. The dashboard displays model performance using accuracy, precision, recall, F1-score, and a confusion matrix. It also allows users to enter student details and predict whether a student is At Risk or Performing Well. The goal is to support early identification of students who may need additional academic assistance.

Objective of the Project

The main objective of the Automated Student Performance Prediction project is to develop an AutoML-based system that predicts student academic performance and identifies students who may be at risk of poor academic outcomes. The system aims to automate data preprocessing, model training, hyperparameter tuning, and model selection by comparing multiple machine learning algorithms. It evaluates the selected model using accuracy, precision, recall, and F1-score. An interactive dashboard allows users to enter student details, view predictions, and understand the factors influencing performance. The ultimate goal is to support early intervention and help educators provide timely academic assistance to students who need it.

Expected Outcome of the Project

1. Develop an AutoML system that automatically compares machine learning algorithms and selects a suitable model for student performance prediction.

2. Identify students who may be academically at risk using student-related attributes and display predictions through an interactive dashboard.

3. Evaluate model performance using accuracy, precision, recall, F1-score, and a confusion matrix to support early identification and timely academic intervention.

Resources Required

- **List of hardware components**
  - Laptop or computer with 8 GB RAM or higher
  - Internet connection
  - Standard processor (Intel i5 / AMD Ryzen 5 or equivalent)

- **List of software tools**
  - Python
  - Jupyter Notebook / Google Colab
  - VS Code
  - Streamlit
  - Scikit-learn
  - Pandas and NumPy
  - Matplotlib and Seaborn